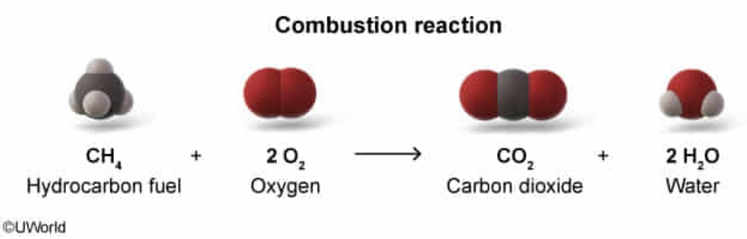
Which of the following is a combustion reaction?

- A. $H_2CO_3 \rightarrow CO_2 + H_2O$ [11%]
- B. $C + H_2O \rightarrow CO + H_2$ [1%]
- C. $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ [84%]
- D. $C + CO_2 \rightarrow 2\ CO$ [1%]

Correct
85% answered correctly

Explanation:



Chemical reactions can be categorized broadly (but not exclusively) into five main types: combination, decomposition, single replacement, double replacement, and **combustion**. Combustion reactions involve the burning of a **fuel** (often a hydrocarbon, an alcohol, or another flammable organic compound) in the presence of **oxygen**, often from the air. When the supply of oxygen is plentiful, complete combustion occurs to form **carbon dioxide (CO_2)** and **water**. The reaction of methane (CH_4), a hydrocarbon fuel, with oxygen to form CO_2 and water is an example of a complete combustion reaction.

Variations of combustion reactions also exist. For example, if the supply of oxygen is limited, carbon monoxide (CO) may form. Additionally, if the fuel contains atoms of nitrogen, sulfur, or other elements, then **oxides** of these atoms can also be produced as products.

(Choice A) Although CO_2 and H_2O are the typical products formed during combustion, $H_2CO_3 \rightarrow CO_2 + H_2O$ is a decomposition reaction, wherein *one* reactant species breaks down into two or more product species (the opposite pattern of combination reactions). No fuel is being burned in the presence of oxygen in this reaction.

(Choice B) The reaction $C + H_2O \rightarrow CO + H_2$ follows the pattern of a single replacement, wherein a compound reacts with a free *element* (a substance in its elemental form that is not bonded with other types of atoms) and forms a new compound and a new *element* as the products.

(Choice D) The reaction $C + CO_2 \rightarrow 2\ CO$ is a combination reaction, wherein two or more chemical species combine and form *one* product species.

Educational objective:
Chemical reactions can be categorized broadly into five main types: combination, decomposition, single replacement, double replacement, and combustion. Combustion reactions involve the burning of a fuel in the presence of oxygen to form carbon dioxide and water (and sometimes other oxides).

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